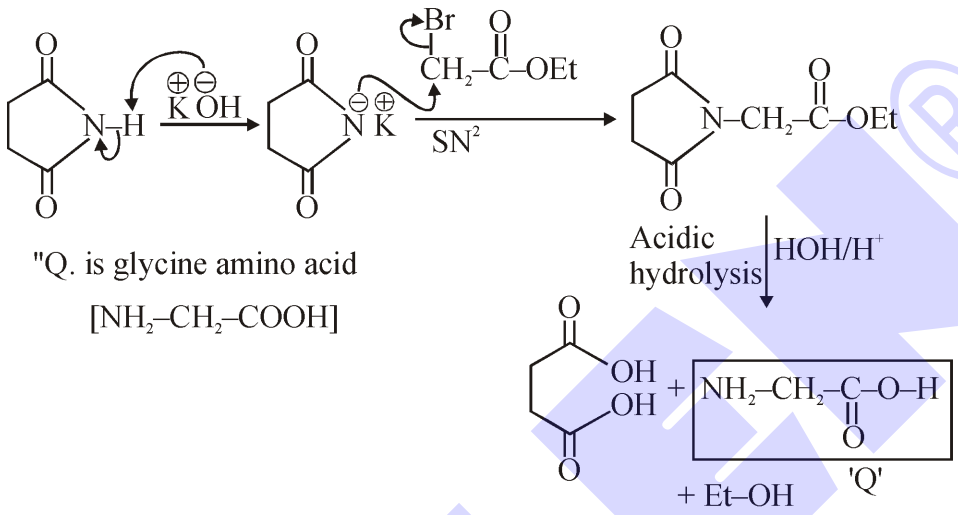
SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

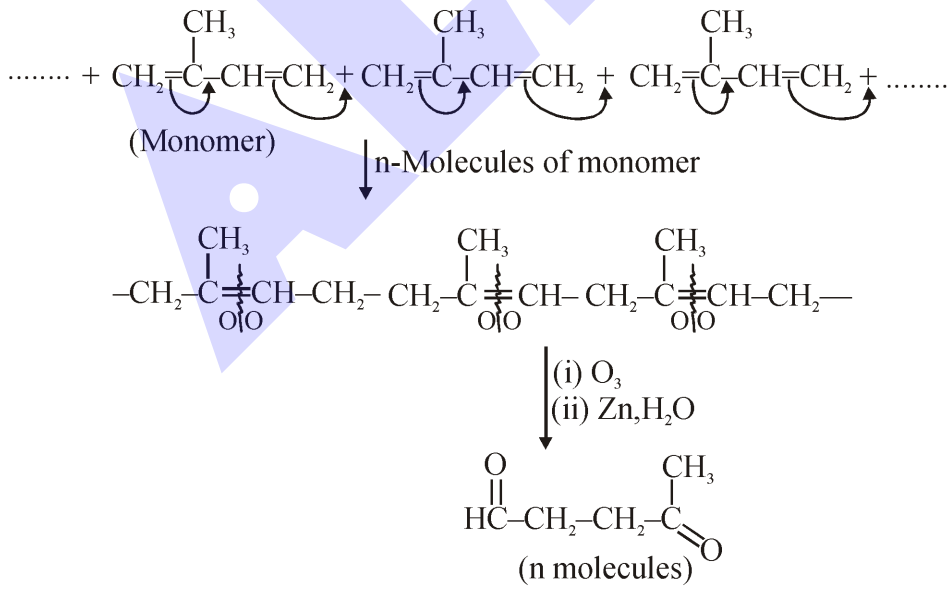
1. (C)

Sol.



2. (C)

Sol.



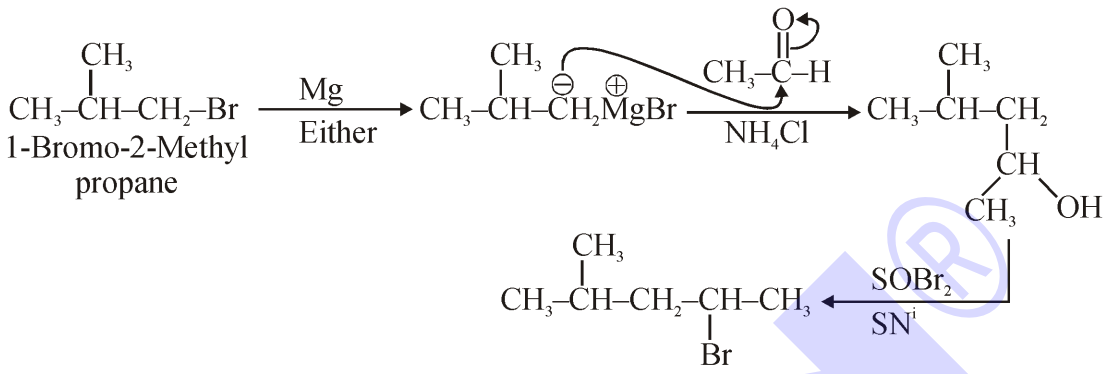
For More Material Join: @JEEAdvanced_2025

3. (A)

Sol. Aniline is steam volatile and B.P. of aniline is 184 C^{o} , So impurities are easily separated by steam distillation, and aniline is finally easily separated from water by decreasing temperature.

4. (A)

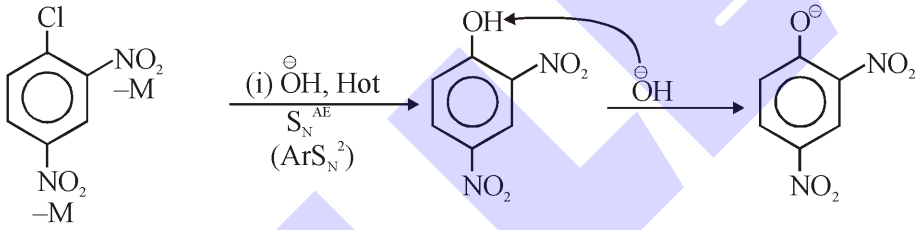
Sol.



2-Bromo-4-Methyl pentane

5. (D)

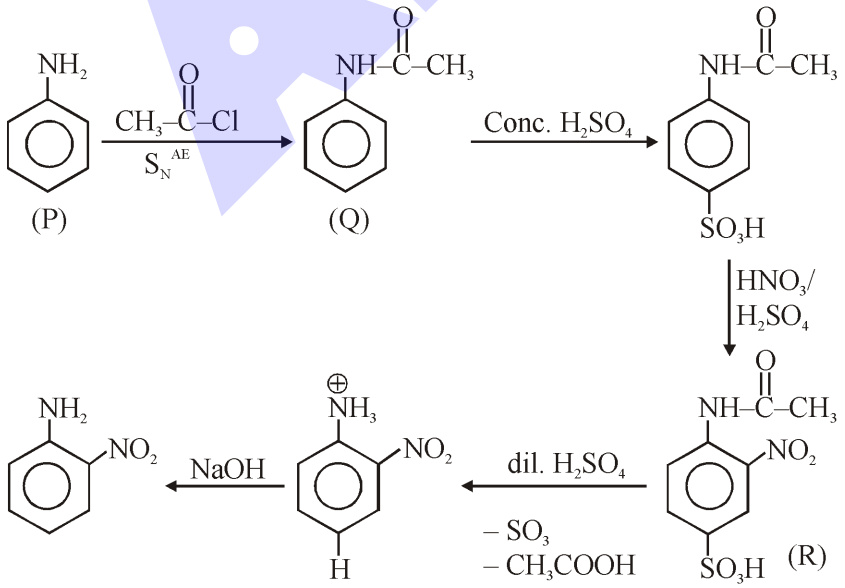
Sol. Compound A gives ArSN^{2}



Compound 'B' and compound 'C' gives NAR (Nucleophilic addition reaction)

6. (D)

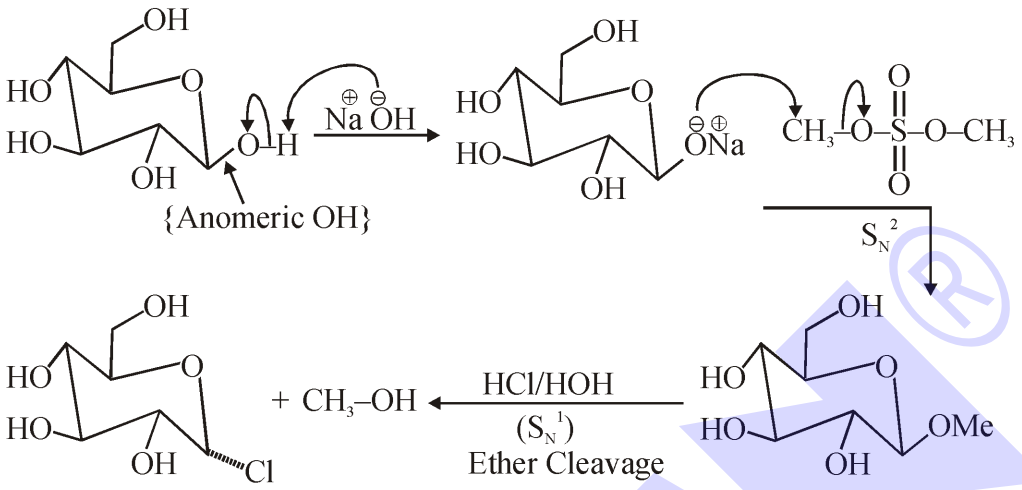
Sol.



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7. (B,C,D)

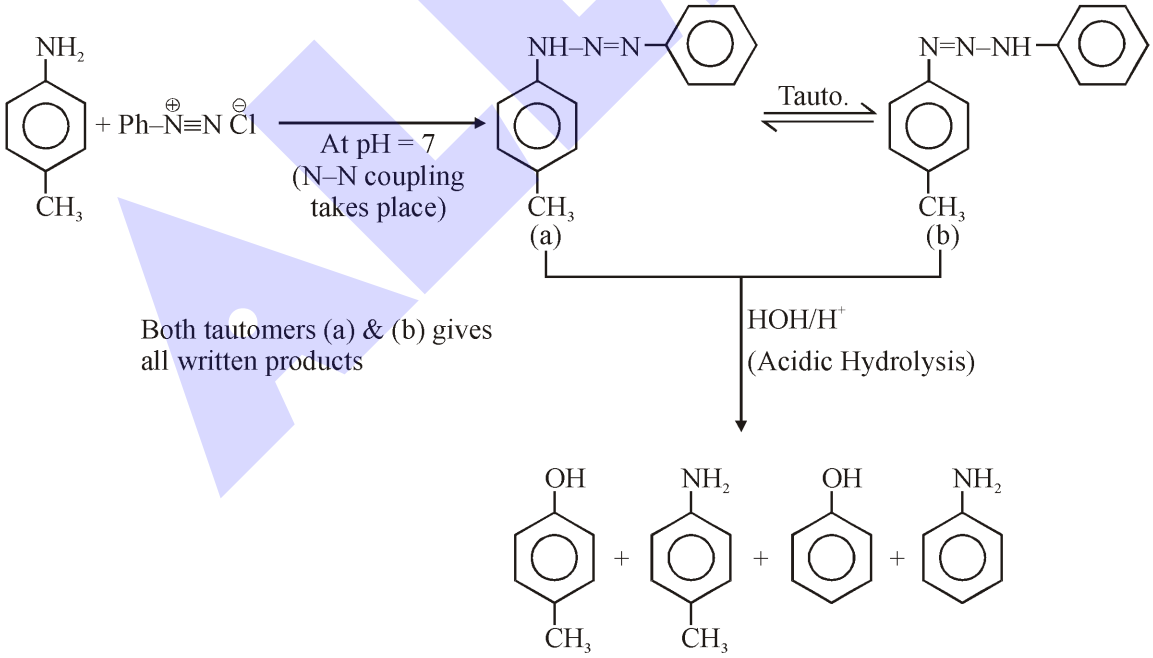
Sol. Product of 'A' is not correctly matched



B,C,D products are correctly matched .

8. (A,B,C,D)

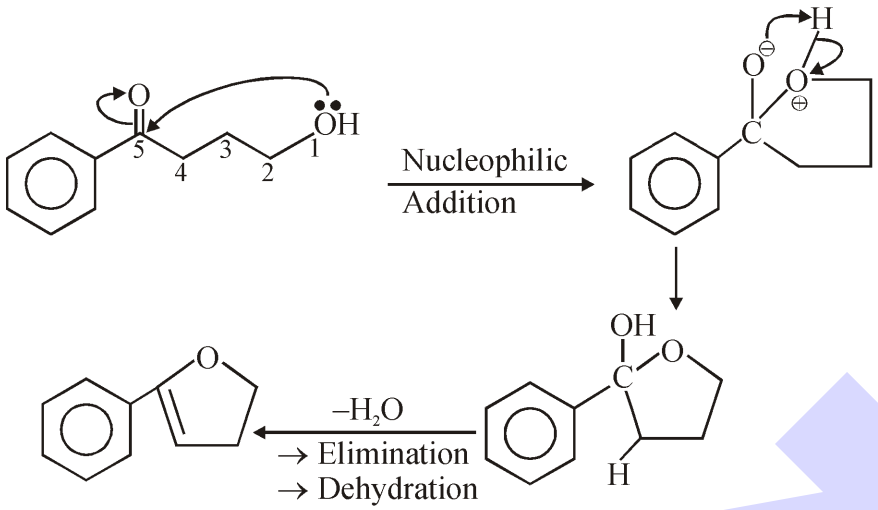
Sol.



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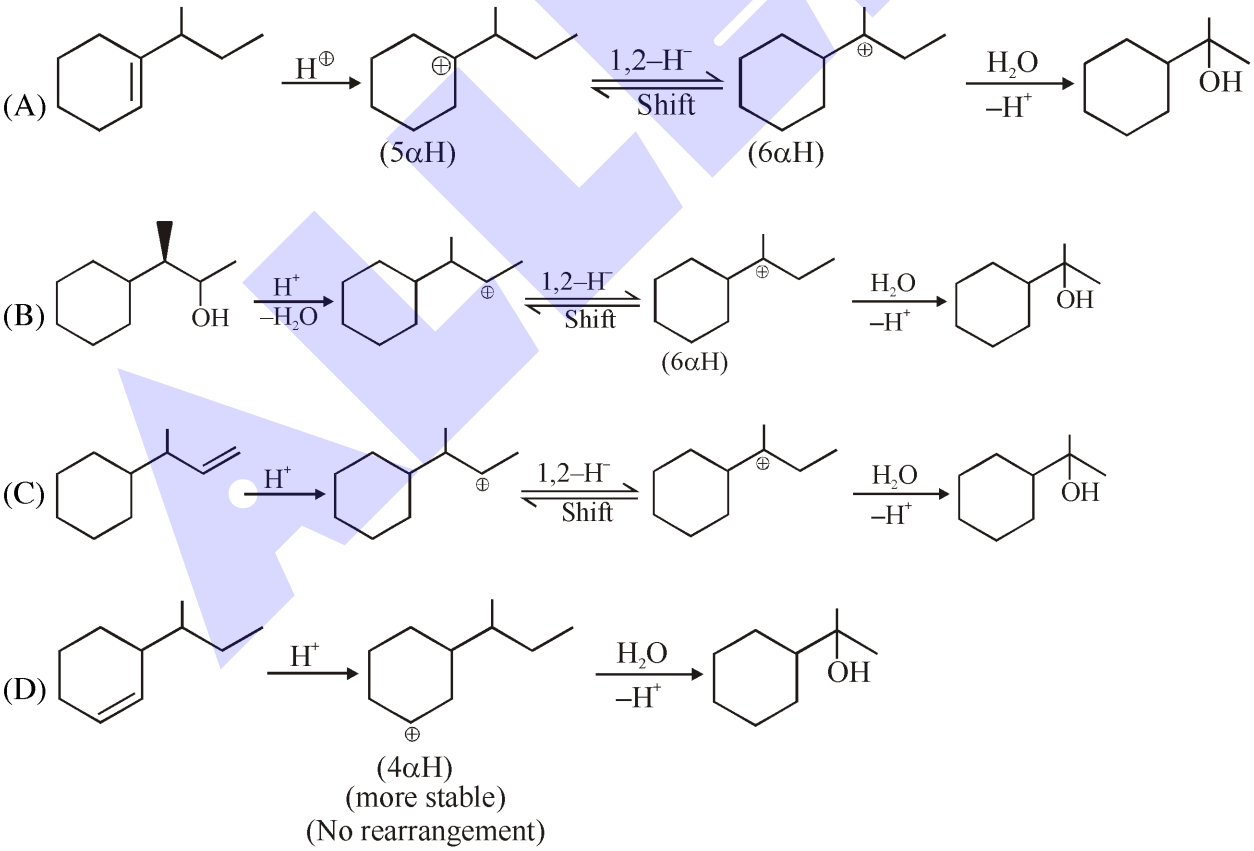
9. (B,C,D)

Sol.



10. **(A,B,C)**

Sol.



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11. (A,B,C,D)

12. (B,D)

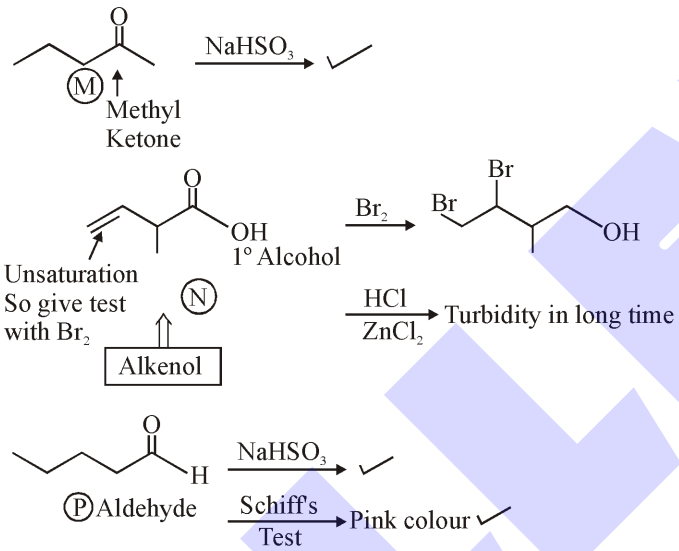
Sol. * Only aldehyde and methyl ketones give test with $NaHSO_3$

* Turbidity after prolong heating means 1° alcohol is present.

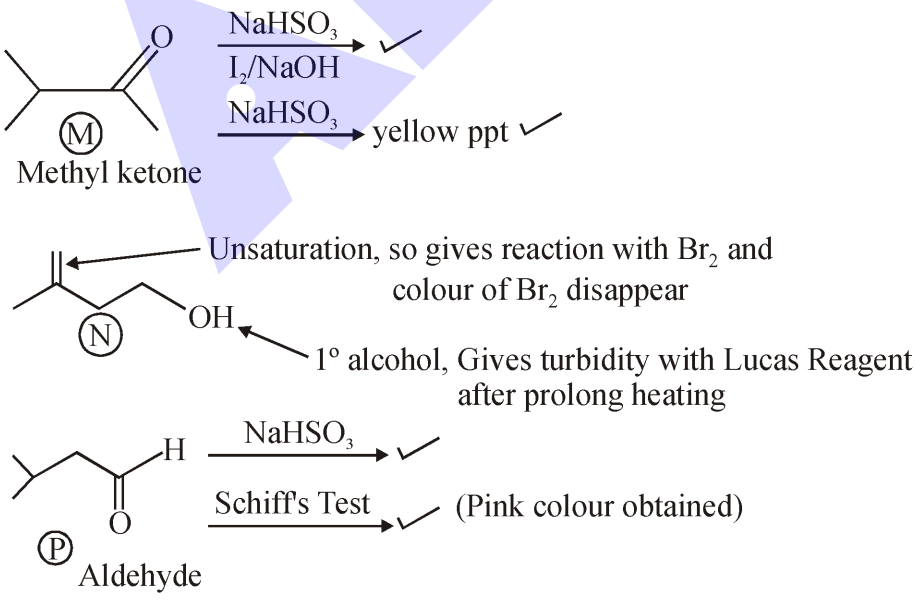
* Generally, $CH_3-\underset{\underset{OH}{|}}{CH}-R$ or $CH_3-\underset{\underset{O}{||}}{C}-R$ gives Iodoform test,

* Aldehyde gives Schiff's test.
(give pink colour with p-rosaniline hydrochloride)

Option-B



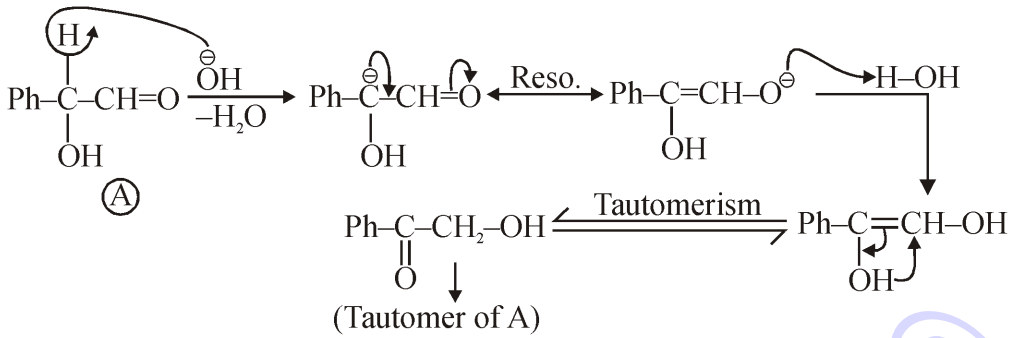
Option-D



For More Material Join: @JEEAdvanced_2025

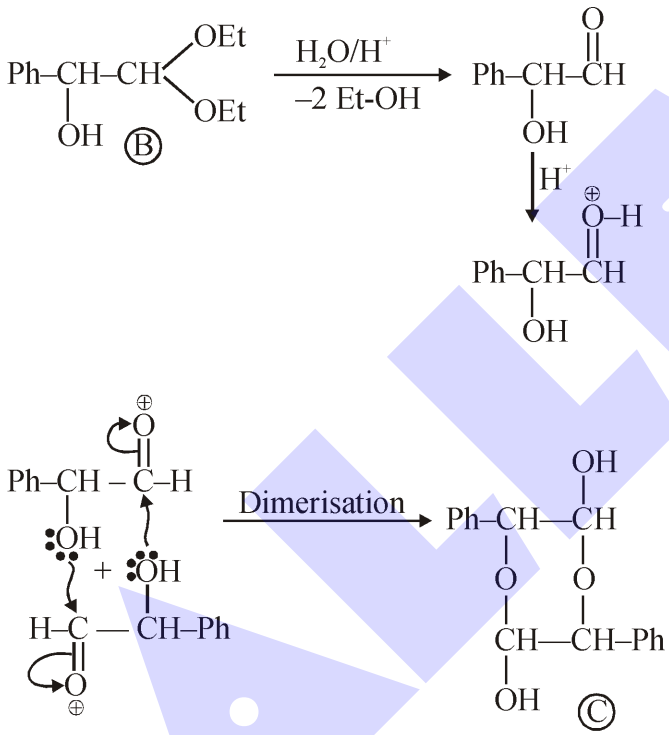
13. (C)

Sol.



14. (B)

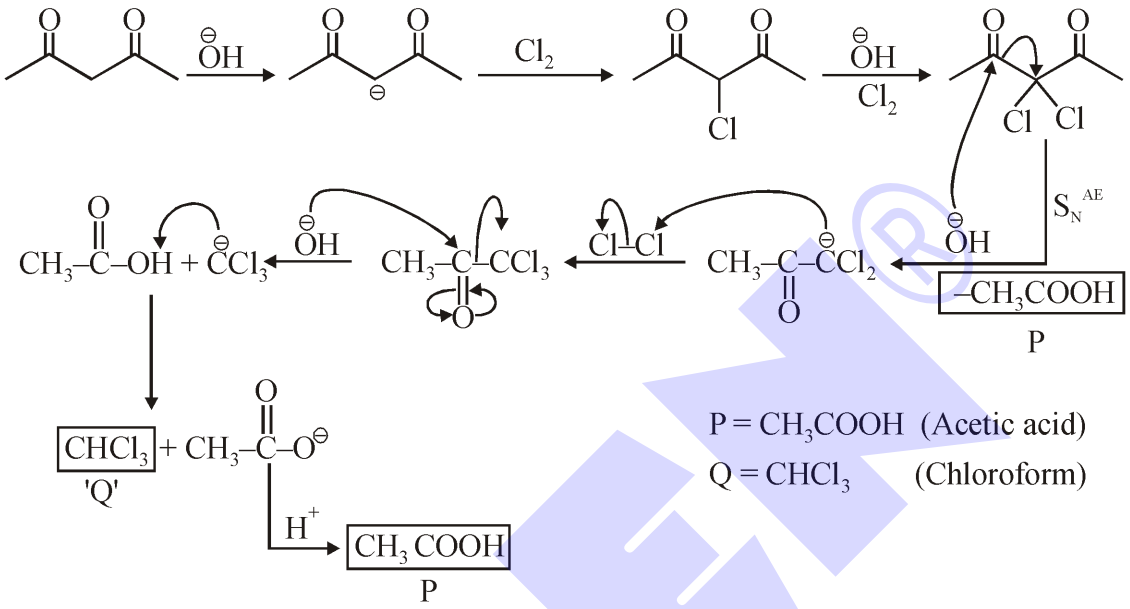
Sol.



For More Material Join: @JEEAdvanced_2025

15. P → 3, Q → 4, R → 2, S → 1

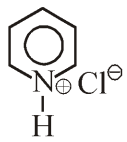
Sol. (i)



P = CH_3COOH (Acetic acid)

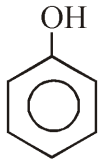
Q = $CHCl_3$ (Chloroform)

(ii) $CH_3-\overset{O}{\overset{\|}{C}}-OH$ (P) $\xrightarrow[-HCl]{Py}$ $CH_3-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3$ (Acetic anhydride) (R) + Pyridinium chloride

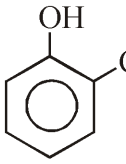


(iii) $Ph-CHO + CH_3COOH \xrightarrow[(ii)\ H_3O^+,\Delta]{(i)\ (CH_3CO)_2O}$ (perkin Reaction) $Ph-CH=CH-COOH$ (T) (Cinnamic acid)

(iv)



+ $CHCl_3$ $\xrightarrow[(ii)\ H^+]{(i)\ KOH}$ [R.T.R]

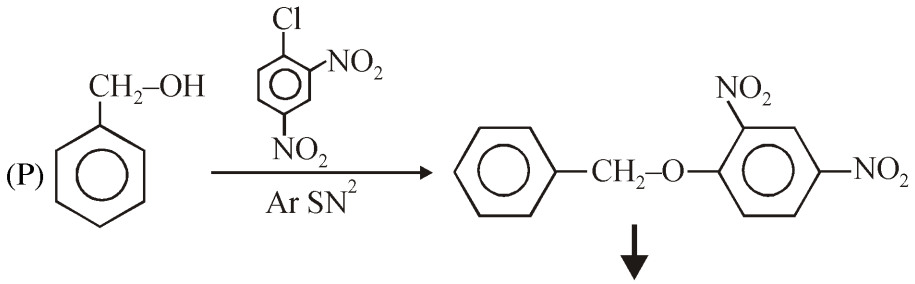


(Salicylaldehyde)

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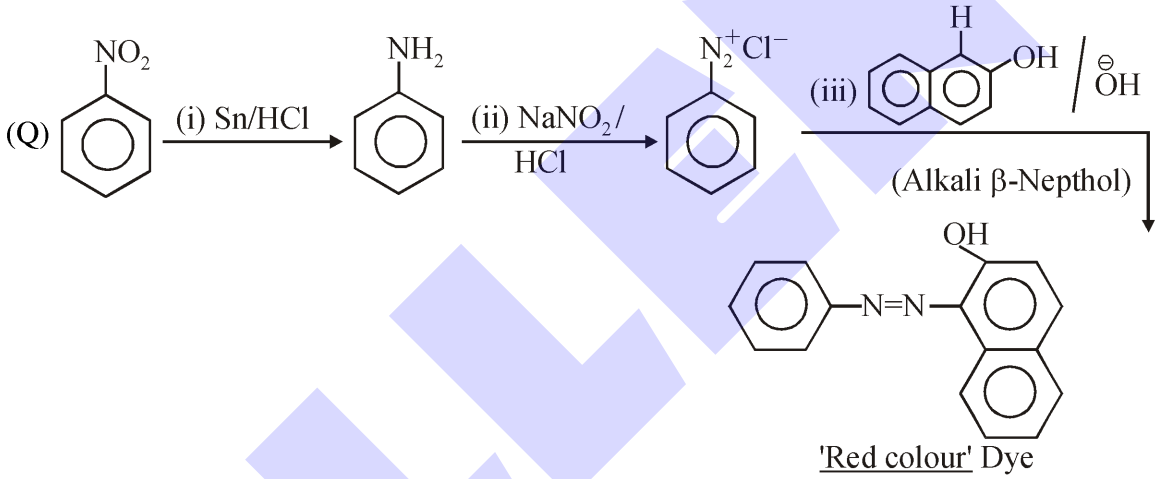
16. **P → 3, Q → 2,3, R → 1, S → 3,4**

Sol. (P)



"Since N is present in compound, so it's Lassaigne's solution will give blue colour with $FeSO_4$ solution.

(Q)



→ Due to presence of phenolic –OH, it will give violet colour with neutral $FeCl_3$ solution.

→ Due to presence of N, Lassaigne solution of formed compound will give blue colour with $FeSO_4$ solution.

(R) η-Glucose $\xrightarrow[\text{Polymerisation}]{H^+}$ Starch

↓

It will give 'Molish Test'

(S) Amino Acid Mixture $\xrightarrow{\text{Polymerisation}}$ Protein

→ Amino acid gives 'Biurate test'

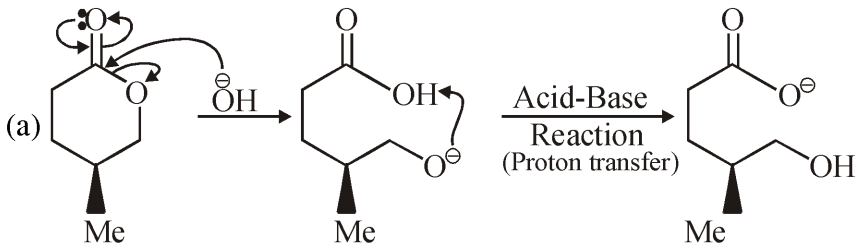
→ Due to presence of N, it's L.S. will give blue colour with $FeSO_4$ Solution

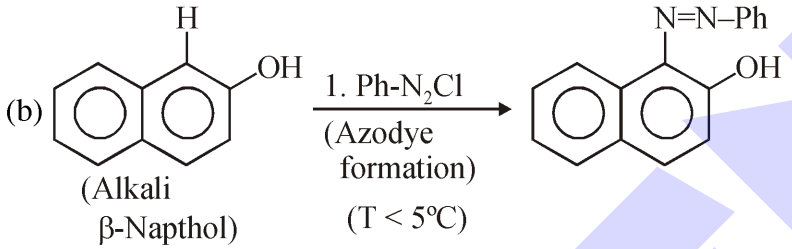
For More Material Join: @JEEAdvanced_2025

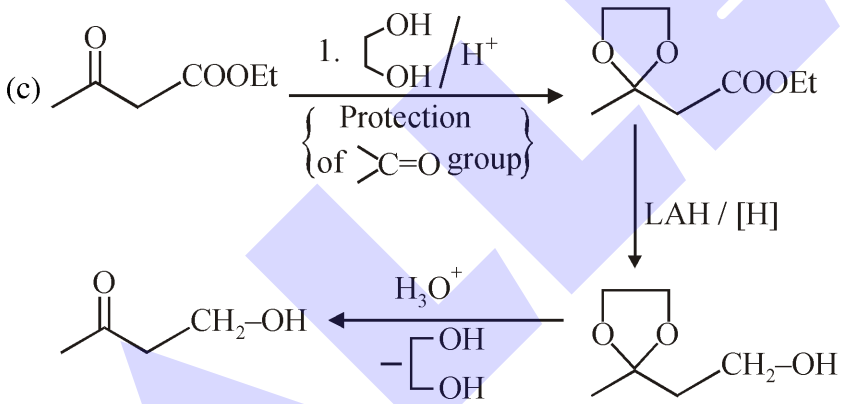
SECTION-II (Integer Type)

1. (4)

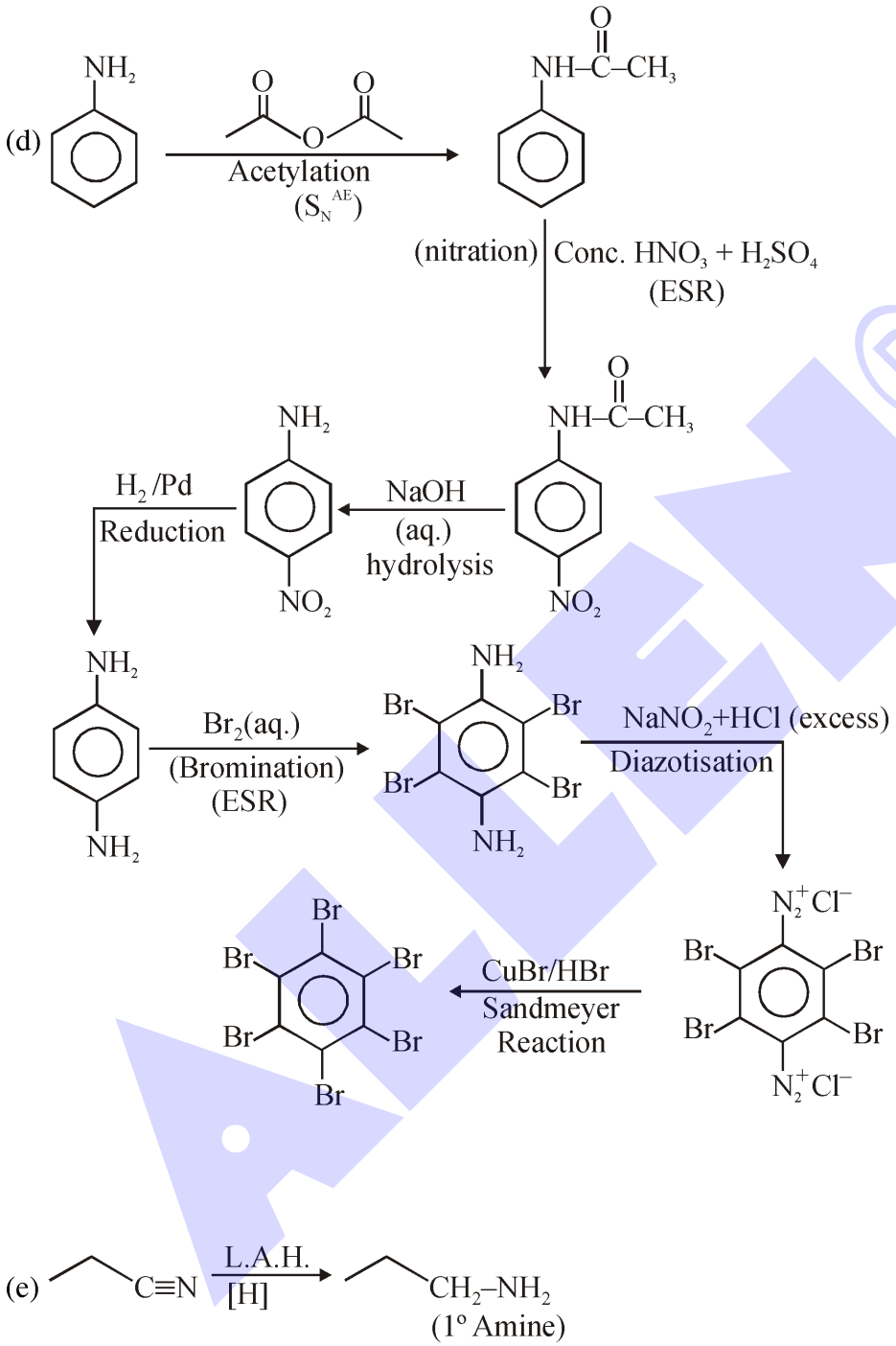
Sol. (a) ✓, (b) ✓, (c) ✓, (d) ✓, (e) ×, (f) ×



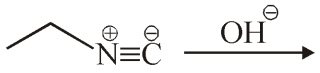




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(f)  isocyanide do not undergo hydrolysis in basic medium.

2. (8)

Sol. **(a) ✓, (b) ×, (c) ✓, (d) ×, (e) ✓, (f) ✓, (g) ✓, (h) ✓, (i) ✓, (j) ✓**

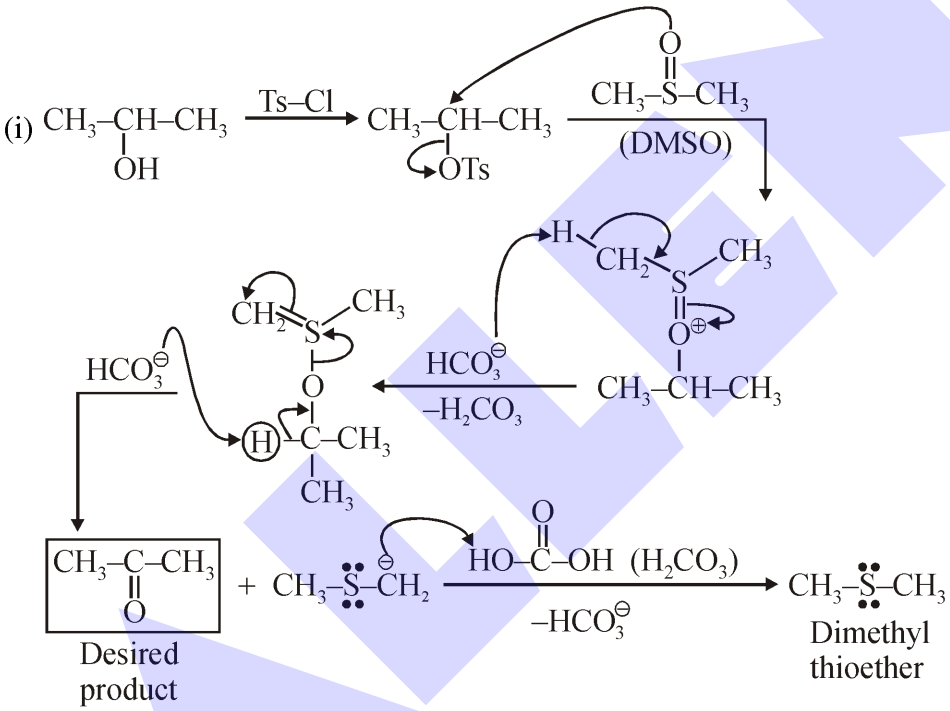
(c) NBS → N-Bromo succinimide

→ Oxidises alcohol into Aldehyde or ketone

(g) Oppenaur oxidation

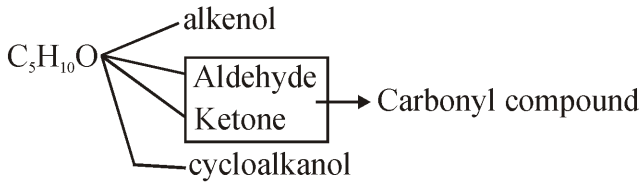
(h) Collin's Reagent → Moderate oxidising agent

→ [Py + CrO_3 + CH_2Cl_2]

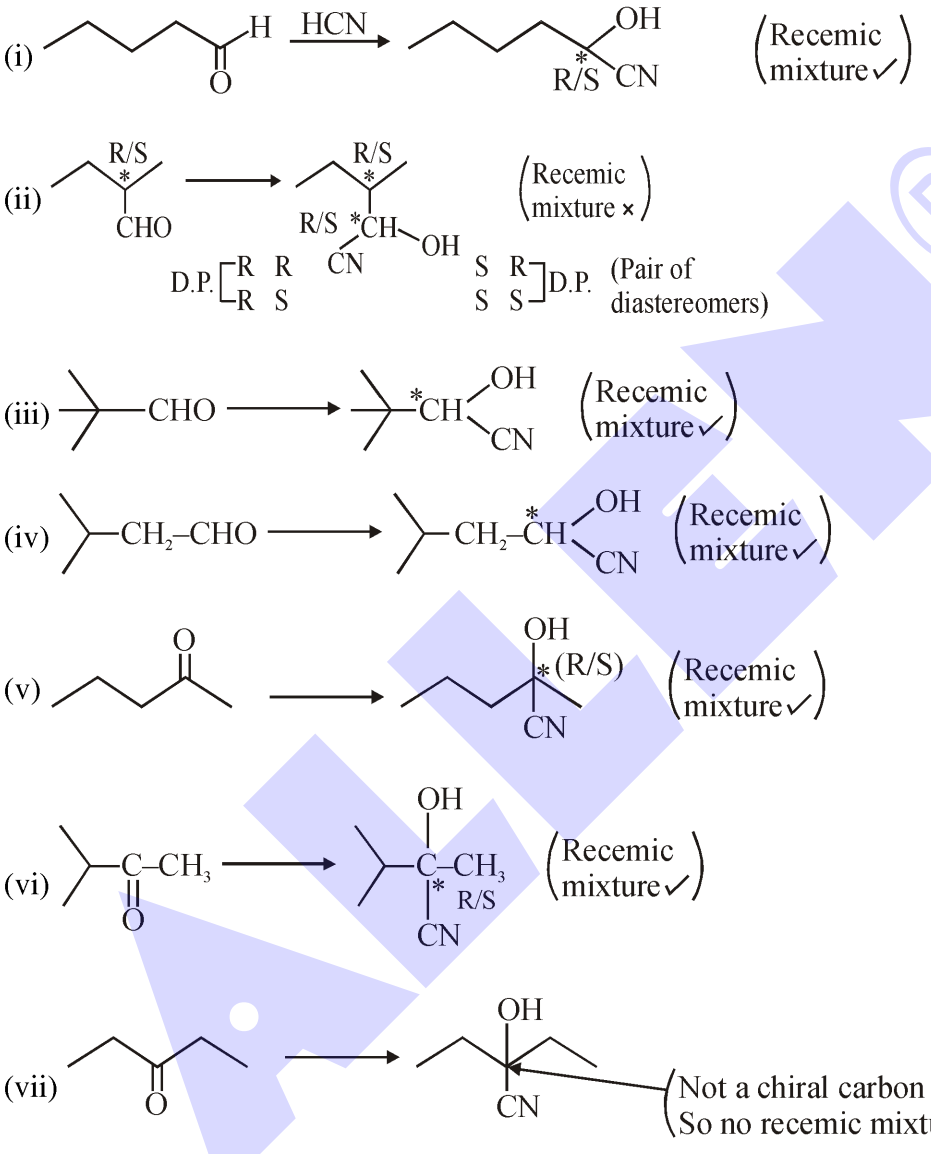


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3.

Sol. 

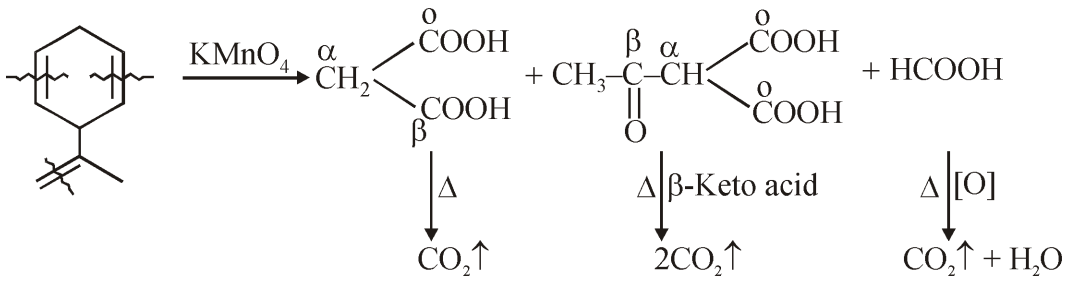
(Ans. 5)



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4.

Sol.



So total moles of CO_2 evolved $\Rightarrow 4$

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